

CHAPTER 4 :- Exploring Magnets

Topic 1: Magnets and Magnetic Materials

Definition

A magnet is an object that attracts certain specific materials. Materials attracted by a magnet are called **magnetic materials**, while those not attracted are **non-magnetic materials**.

Concept Explanation

Historically, magnets were discovered as naturally occurring rocks called lodestones. Today, we manufacture artificial magnets in various shapes to suit different needs, such as bar magnets, U-shaped magnets, ring magnets, and cylindrical magnets.

A magnet does not attract everything. It acts like a picky selector. It will powerfully attract metals like Iron, Nickel, and Cobalt (magnetic materials). However, it completely ignores materials like wood, plastic, rubber, glass, and paper (non-magnetic materials).

Diagram



Real-Life Example

The doors of refrigerators and the lids of many pencil boxes shut tightly on their own because they have hidden magnetic strips inside them that attract the iron in the frame.

☑ Quick Revision Checklist:

- [] Lodestones are natural magnets; bar/U-shape are artificial.
- [] Magnetic materials: Iron, Nickel, Cobalt.
- [] Non-magnetic materials: Wood, glass, plastic, rubber.

Topic 2: Poles of a Magnet

Definition

Magnetic poles are the specific regions at the two ends of a magnet where the magnetic pulling force is the absolute strongest.

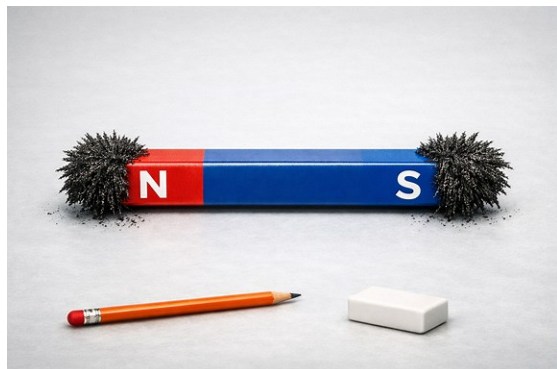
Concept Explanation

No matter the shape of the magnet, its power is not spread out evenly. The attractive force is concentrated at the two ends, which are called the **North Pole (N)** and the **South Pole (S)**.

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A fundamental rule of nature is that magnetic poles *always* exist in pairs. It is impossible to isolate a single pole. If you take a bar magnet and break it perfectly in half, you do not get one isolated North piece and one isolated South piece. Instead, new poles instantly form at the broken ends, giving you two smaller, complete magnets!

Diagram



Real-Life Example

If you accidentally drop a brittle bar magnet and it snaps into three pieces, you haven't destroyed the poles. You now just have three smaller, complete magnets, each with its own North and South pole.

☑ Quick Revision Checklist:

- [] The maximum magnetic force is at the poles.
- [] Every magnet has a North (N) and a South (S) pole.
- [] A single pole cannot exist; poles always come in pairs.

Topic 3: Finding Directions (The Magnetic Compass)

Definition

A magnetic compass is a navigational instrument containing a freely rotating magnetic needle that automatically aligns itself to point toward the Earth's North-South direction.

Concept Explanation

If you tie a thread to the center of a bar magnet and let it hang freely, it will always stop and rest in the exact same direction: North-South. Why does this happen? Because the Earth itself behaves like a giant magnet, and its magnetic field pulls on the suspended magnet!

Sailors and travellers used this property to navigate. This led to the invention of the magnetic compass. It consists of a tiny magnetic needle inside a small box with a dial. The tip of the needle (usually painted red) is the North-seeking pole. By rotating the dial to align with the needle, you can figure out all other directions (East, West, South).

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Diagram



Real-Life Example

Before GPS existed on smartphones, sailors in the middle of the ocean used magnetic compasses to find their way during heavy storms when the guiding stars were hidden by clouds.

☑ Quick Revision Checklist:

- ☐ Freely suspended magnets always align North-South.
- ☐ The Earth acts like a giant magnet.
- ☐ The red tip of a compass needle points to the North.
- ☐ You can magnetize an iron needle by repeatedly stroking it in one direction with a permanent magnet.

Topic 4: Attraction and Repulsion Between Magnets

Definition

Attraction is the pulling force between two unlike magnetic poles, while **Repulsion** is the pushing force between two like magnetic poles.

Concept Explanation

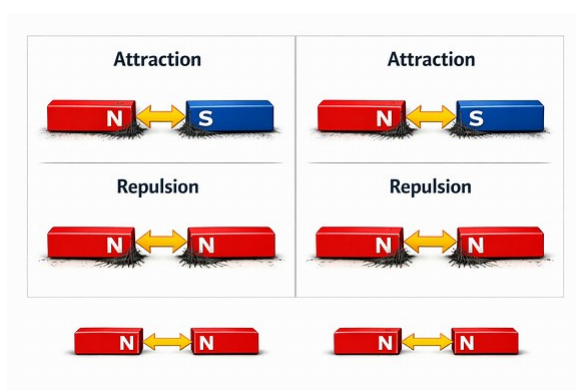
When two magnets meet, they interact following a strict set of rules:

1. **Unlike Poles Attract:** A North pole will strongly pull toward a South pole.
2. **Like Poles Repel:** A North pole will push away another North pole. Similarly, a South pole will repel another South pole.

Interestingly, magnets don't even need to touch to exert these forces. Furthermore, this magnetic force can pass right through non-magnetic materials. If you put a piece of wood, plastic, or glass between a magnet and a compass, the magnetic force will pass right through the barrier and still deflect the compass needle!

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Diagram



Real-Life Example

Maglev (Magnetic Levitation) trains use the power of magnetic *repulsion* to push the train slightly off the tracks. Because it hovers, there is no friction, allowing the train to travel at incredibly high speeds!

☑ Quick Revision Checklist:

- [] Unlike poles (N-S) attract.
- [] Like poles (N-N or S-S) repel.
- [] Repulsion is the surest test to confirm if an object is a magnet (an iron bar will just attract both sides).
- [] Magnetic force easily passes through non-magnetic barriers (wood, glass, paper).

Topic 5: Proper Care and Storage of Magnets

Definition

Magnets can lose their magnetic properties (demagnetization) if mishandled, so they require specific storage techniques using soft iron "keepers" to preserve their strength.

Concept Explanation

Magnets are not permanently invincible. They will lose their magnetic strength if they are:

1. Heated over a fire.
2. Dropped repeatedly from a height.
3. Hammered or beaten with force.

To keep them safe when not in use, bar magnets should be stored in pairs. You must place them side-by-side with *unlike* poles on the same side (N next to S). A piece of wood should separate them down the middle, and two pieces of soft iron (called keepers) should be placed across their ends to complete the magnetic circuit. Lastly, magnets should always be kept away from electronics like mobile phones and remote controls to prevent damage.

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Diagram



Real-Life Example

If a mechanic forcefully hammers a magnetized screwdriver, it will lose its magnetism and will no longer be able to easily pick up dropped steel screws.

☑ Quick Revision Checklist:

- ☐ Never heat, drop, or hammer a magnet.
- ☐ Store bar magnets in pairs with opposite poles facing the same way.
- ☐ Use soft iron 'keepers' across the poles.
- ☐ Keep magnets away from sensitive electronic devices.